

Predicting hERG Channel Blockade

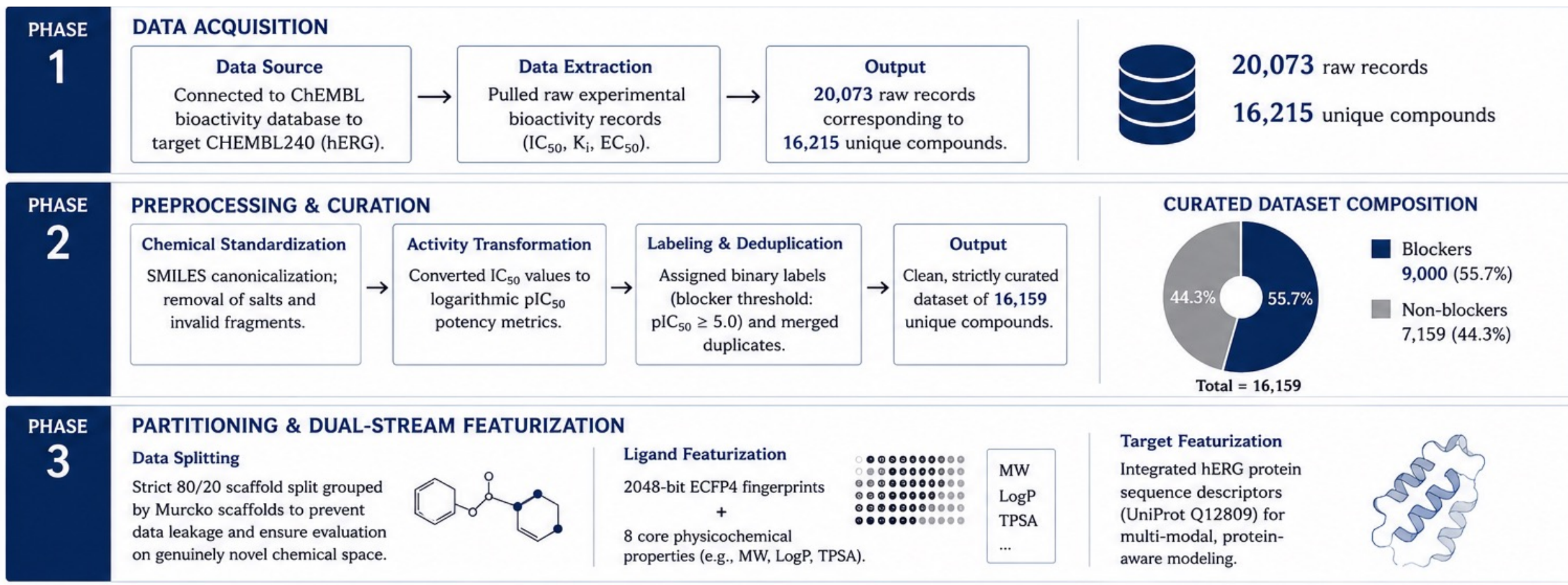
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Team:- CardioSafe

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GitHub Repository - <https://github.com/AdityaChourasia2309/CardioSafe>

Workflow



Workflow

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PREDICTIVE MODELING & INTERPRETABILITY

Classification Benchmarking
Models trained and evaluated on held-out scaffold test set.

- Logistic Regression
- Random Forest
- XGBoost

Classification Results (Random Forest – Best)	
Accuracy	0.777
ROC-AUC	0.865
PR-AUC	0.897

Regression Pipeline (Random Forest Regressor)	
pIC ₅₀ prediction performance	
RMSE	0.688

Interpretability (SHAP Analysis)
Top predictive features identified, consistent with hERG biophysical pharmacology.



Feature	Mean SHAP value
TPSA	~0.85
LogP	~0.75
HBA	~0.55
...	~0.25

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Reliability Mapping (Held-out Scaffold Test Set)

Domain	Definition	Count (n)	Confidence	ROC-AUC	Accuracy
In-Domain	High-confidence predictions for compounds within applicability domain	2,943	High (≥ Threshold)	0.876	0.790
Out-of-Domain	Low-confidence predictions for structurally novel compounds	350	Low (< Threshold)	0.736	0.660

Complete Run Instructions

Pipeline Execution Workflow (Data Acquisition & Curation)

• Execution Environment & Protocol

- Execute python notebooks sequentially from the repository root.
- Update the shared SAVE_DIR variable at the top of each notebook to match your local or drive environment before execution.

• Step 1: Data Acquisition (01_fetch_data.ipynb)

- **Objective:** Extract raw bioactivity records for hERG target ChEMBL240 from the ChEMBL database.
- **Output:** data/raw/raw_chembl_herg.csv

• Step 2: Data Curation and Cleaning (02_curate.ipynb)

- **Objective:** Standardize chemical representations via RDKit SMILES canonicalization, convert IC50 to pIC50, derive binary blocker labels ($\text{pIC50} \geq 5.0$), and de-duplicate records.
- **Verification:** Documents explicit counts at each filtering step to confirm zero silent data loss.
- **Outputs:** data/herg_clean.csv, reports/data_curation_report.md, and distribution plots.

• Step 3: Partitioning and Featurization (03_split_featurize.ipynb)

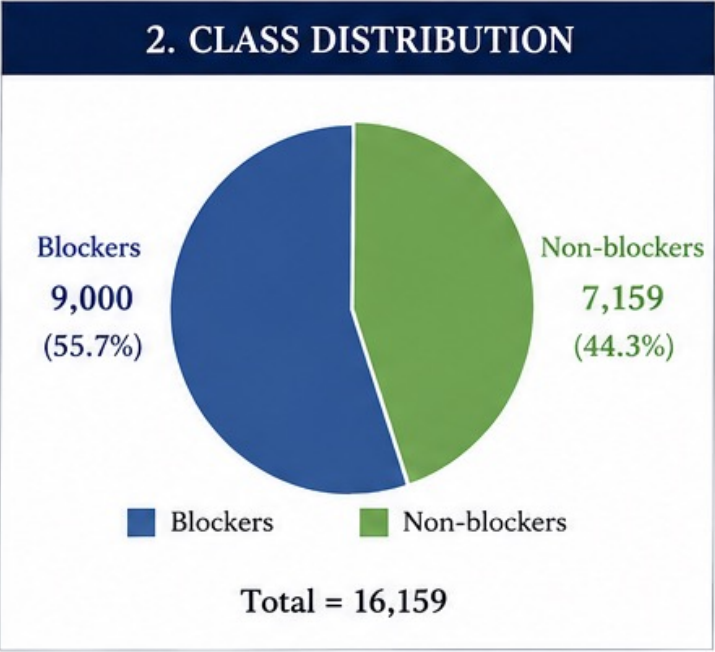
- **Objective:** Generate a strict 80/20 scaffold split using Murcko scaffolds alongside a diagnostic random-split baseline to quantify evaluation bias.
- **Features:** Computes 2048-bit ECFP4 chemical fingerprints and 8 core physicochemical descriptors.
- **Output:** data/herg_split.csv, data/fp_matrix.npy, and data/descriptors.csv.

Complete Run Instructions

- **Step 4: Baseline Model Training and Interpretability (04a & 04b)**
 - **Objective:** Train Logistic Regression, Random Forest (Classification and Regression), and XGBoost models.
 - **Advanced Integration:** Incorporates UniProt Q12809 hERG protein sequence features and runs SHAP interpretability analysis.
 - **Runtime / Outputs:** ~10 to 15 minutes | Checkpoints in checkpoints/, submission.csv, and interpretability plots.
- **Step 5: Applicability Domain and Confidence Scoring (05_applicability_domain.ipynb)**
 - **Objective:** Compute maximum Tanimoto similarity scores against the training set to establish prediction confidence bounds.
 - **Validation:** Validates performance differentials between in-domain and out-of-domain chemical spaces.
 - **Runtime / Outputs:** ~2 to 5 minutes | data/test_with_AD_scores.csv and updated submission files with confidence flags.
- **Inference-Only Deployment (Skipping Retraining)**
 - Pre-trained models can be loaded directly for downstream evaluation using joblib

Results:

1. DATASET COMPOSITION		
Total Unique Compounds		16,159
Dataset Partition		
Subset	Count	Percentage (%)
Training Set	12,866	79.6
Test Set (Scaffold-split)	3,293	20.4
Total	16,159	100.0



3. CLASSIFICATION PERFORMANCE <i>(Random Forest – Molecular Features Only)</i>	
Scaffold-split Test Set (n = 3,293)	
Metric	Score
Accuracy	0.777
F1 Score	0.809
ROC-AUC	0.865
PR-AUC	0.897
<i>All metrics evaluated on held-out scaffold-split test set.</i>	

4. MULTI-MODAL INTEGRATION *(Random Forest with Protein Features)*

Integration of protein features with molecular features yields highly comparable performance.

Performance on Scaffold-split Test Set (n = 3,293)	
Metric	Score
Accuracy	0.776
ROC-AUC	0.866

Performance is highly comparable to the molecular-features-only model.

Results

5. REGRESSION PERFORMANCE (pIC₅₀ PREDICTION)

Random Forest Regressor

Metric	RMSE	MAE	R-squared (R ²)
Score	0.688	0.471	0.486

Performance evaluated on test set for pIC₅₀ potency prediction.

Summary of the models and their respective results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.734	0.759	0.782	0.770	0.810	0.847
Random Forest (best)	0.777	0.788	0.831	0.809	0.865	0.897
RF + Protein Sequence	0.776	0.784	0.839	0.810	0.866	0.896
XGBoost	0.768	0.784	0.818	0.801	0.854	0.883

Results: Split Strategy, Scaffold Integrity and Domain Assessment

1 SPLIT STRATEGY COMPARISON

Split Strategy	Accuracy	ROC-AUC
Random Split <i>(Easier)</i>	0.814	0.900
Scaffold Split	0.777	0.865

2 PERFORMANCE BY APPLICABILITY DOMAIN

Confidence validation testing confirmed the system's reliability across differing structural boundaries.

3 PERFORMANCE BY APPLICABILITY DOMAIN

Domain	Definition	Count (n)	Confidence	Accuracy	ROC-AUC
In-Domain	High-confidence predictions within the model's applicability space	2,943	High (≥ Threshold)	0.790	0.876
Out-of-Domain	Low-confidence predictions for structurally novel compounds outside the applicability space	350	Low (< Threshold)	0.660	0.736

Thank You!